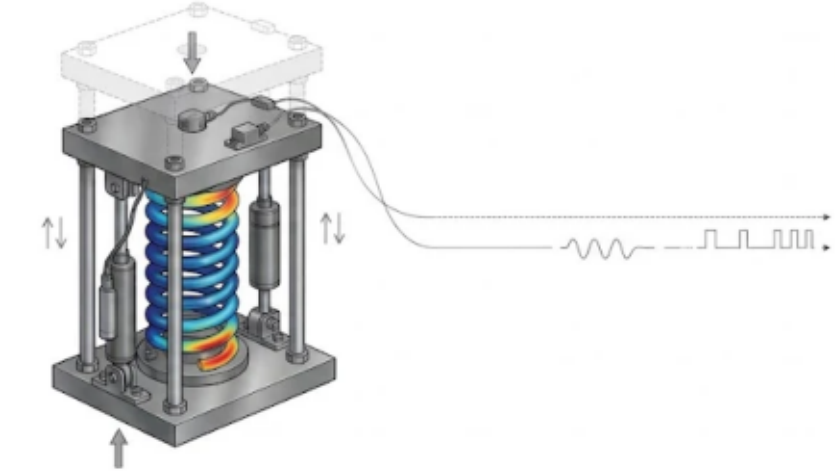
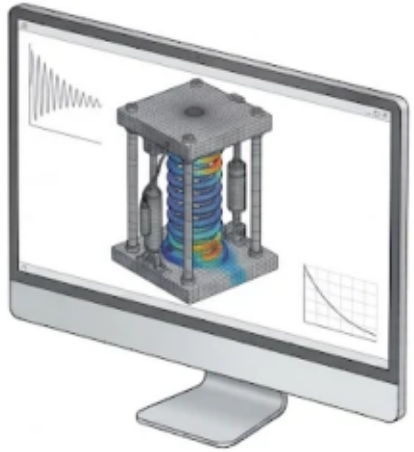
(a) Physical asset + telemetry



(b) Computational twin



(c) What the twin contributes

Learned twin only

self-labelled transitions → reuses the fitted dynamics

Twin + known degradation model

physics-labelled transitions → carries external physics into training data

same downstream identifier

Synthetic trajectories are the transport format, not the source of the physics.